



Research papers

Determination of water required to recover from hydrological drought: Perspective from drought propagation and non-standardized indices

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ABSTRACT

When a severe drought occurs, people often question the amount of water required to recover from the drought which is difficult to determine. Here, we present a method to determine the amount of water required, including both runoff and precipitation, for hydrological drought recovery from a drought propagation perspective. We also consider multi-scale meteorological and hydrological anomaly indices (*MAI* and *HAI*) with non-standardized properties. We present three rainfall-driven basins in southern China to validate the proposed approach using long-term monthly runoff and precipitation records. Drought propagation characteristics (such as duration, severity, development, and recovery processes) were extracted using the *HAI* and *MAI* obtained via a threshold level method and run theory. The proposed method allows estimating the amount of water required, including runoff and the corresponding precipitation, for any hydrological drought to recover to normal using *HAI* and *MAI*. Moreover, a linear relationship between runoff and the corresponding precipitation was established ($R^2 > 0.99$) to estimate the amount of precipitation required to recover from a certain hydrological drought. The methods presented in this study can be used for assessing and managing water resources and ecosystem response during periods of hydrological drought.

1. Introduction

Persistent droughts put substantial pressure on water resources and ecosystems (AghaKouchak et al., 2015a; Belesova et al., 2019; Mishra & Singh, 2010; Sutanto et al., 2020). Droughts can be broadly classified into four types: meteorological, agricultural, hydrological, and socio-economic droughts; these categories refer to the deficits of precipitation, soil moisture, river discharge, and social water demand, respectively (Mishra et al., 2015; Mishra and Singh, 2010). Hydrological drought, characterized by streamflow (or groundwater level) deficit, is driven by continuation (or propagation) of a meteorological drought (Linsley et al., 1982). The impact of droughts typically increases when a meteorological drought turns into a hydrological drought, affecting the ecosystem (e.g., water quality), (Attrill & Power, 2000; Ahmadi, et al., 2019; Li et al., 2018) residents, as well as industrial and agricultural

production (AghaKouchak et al., 2015a; Sheffield et al., 2012; Campana et al., 2018; Zhang et al., 2019). Therefore, water resource managers require drought recovery metrics to monitor droughts and their propagation for managing water resource systems (Borgomeo et al., 2015; Garcia et al., 2019; Kuriqi et al., 2019; Li et al., 2019; Tran et al., 2017).

Following its onset, the lifecycle of a hydrological drought includes development and recovery or termination (Parry et al., 2016a,b). Development is defined as the period from the onset of hydrological drought to the time when the maximum intensity is realized, while the recovery refers to the period from the maximum intensity of hydrological drought to complete recovery (Wu et al., 2018a, 2019). In recent years, the hydrological drought recovery process has received widespread attention. Using the Gravity Recovery and Climate Experiment – Total Water Storage dataset (GRACE-TWS) and extended GRACE-TWS data, for example, the average recovery time of hydrological droughts

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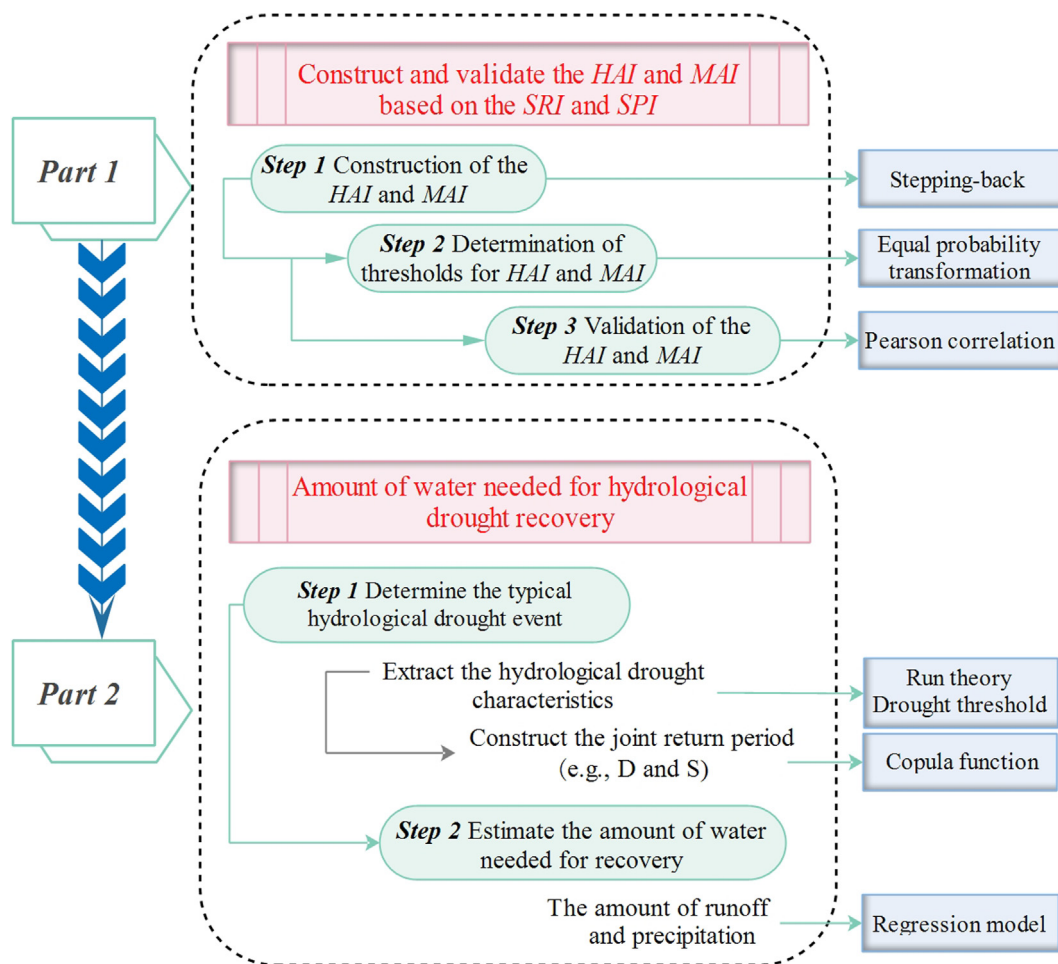


Fig. 1. Flowchart of the proposed framework.

in Texas (Thomas et al., 2014) and the Yangtze River Basin (Zhang et al., 2016) were estimated to be approximately 3–13 and 3.3 months, respectively. Investigating the relationship between hydrological drought recovery time and catchment characteristics (such as elevation, storage, and land cover) of the 52 watersheds in Britain revealed that storage capacity is crucial to realize recovery (Parry et al., 2016a). Wu et al. (2018a) further proposed a systematic approach for describing drought development and recovery characteristics (such as time and severity) and a method for modeling drought propagation speed. Although these previous studies provide valuable information regarding the evolution of hydrological droughts and their recovery times, the amount of water required (e.g., through precipitation) for hydrological drought recovery has not been addressed.

Previous studies have analyzed the precipitation required for meteorological drought recovery (Antofie et al., 2015; Pan et al., 2013). The studies were either based on precipitation shortage during the months preceding drought termination or a particular meteorological drought level value to judge the precipitation required for drought recovery. However, hydrological drought recovery also includes the gradual accumulation of water in streams and rivers to return to normal, rather than being dependent solely on the amount of precipitation (AP) in the preceding months. Thus, it is worth investigating the water required for hydrological drought recovery from a propagation perspective. Note that the term 'water' henceforth refers to both the amount of runoff and its corresponding precipitation required for hydrological drought recovery.

There are several drought indices designed for drought monitoring, prediction, and water resources management purposes (Mishra & Singh, 2010; Shukla et al., 2011). The Standardized Streamflow Index (SSI)

(Vicente-Serrano et al., 2011) and the Standardized Runoff Index (SRI) (Shukla & Wood, 2008) evolved from the Standardized Precipitation Index (SPI) (widely used for identifying meteorological drought) (McKee et al., 1993). Thus, SSI and SRI share the advantages of SPI (i.e., multiple timescales, simple calculation, and less data input). However, they do not directly reflect the amount of water required for recovery as they are dimensionless indices. Streamflow Anomaly Percentage (SAP) (Zaidman et al., 2002) and Streamflow Anomaly Index (SAI) (Wang & Feng, 2000) maintain dimensionality properties. However, they employ a fixed timescale (e.g., monthly or annual) and average streamflow as the calculation parameter; consequently, the cumulative effects of drought for a prolonged event are neglected (Mishra & Singh, 2010). Fixed or variable thresholds based on percentiles of the flow duration curve (e.g., thresholds between the 70th and 90th percentile) are also frequently applied to extract hydrological drought characteristics (i.e., duration and severity) with non-standardized indices (Van Loon et al., 2016). Overall, droughts have multiple timescale features (e.g., 1-month, 3-month, or 12-month timescales) (Mishra & Singh, 2010; Mo & Lettenmaier, 2018). Thus, a non-standardized dimensional index with multiple timescales is required to identify streamflow shortage and the amount of water required for hydrological drought recovery.

Therefore, this study aims to obtain the amount of water required for hydrological drought recovery based on a non-standardized dimensional index with multiple timescales. We performed three main tasks:

- Constructing reliable meteorological and hydrological anomaly indices (MAI and HAI) with non-standardization and multiple timescales features.

- Establishing a method to obtain the amount of water required for hydrological drought recovery from the propagation perspective with the *MAI* and *HAI*.
- Revealing the relationship between the amounts of runoff required and the corresponding precipitation.

This paper presents a simple and convenient method for drought identification and estimation of the water required for hydrological drought recovery.

2. Methodology

2.1. Overview

The methodological framework proposed in this paper includes two parts (Fig. 1). Part 1: Construct and validate the *HAI* and *MAI*, and Part 2: Determine the amount of water required for hydrological drought recovery. The detailed methods for the construction and validation of *HAI* and *MAI* are presented in Section 2.2.2 (Part 1 in Fig. 1). Note that the method for *HAI* and *MAI* construction is consistent with *SPI* and *SRI*, which are two widely used standardized meteorological and hydrological drought indices (Mishra & Singh, 2010). Section 2.2.3 (Part 2 in Fig. 1) introduces the steps to obtain the amount of water required and corresponding precipitation for recovering hydrological drought based on the *MAI* and *HAI* indices.

2.2. Hydrological and meteorological anomaly indices (Part 1)

2.2.1. Construction of *HAI* and *MAI* (Step 1 in Part 1)

The long-term runoff and precipitation records for a certain basin were separated in a 2:3 ratio, in which the first 66.66% series were used to construct the *HAI* and *MAI* series, and the remaining data were used to validate the rationality of the obtained *HAI* and *MAI* series. The anomaly of regional runoff during a certain period compared with the average regional runoff obtained over a long period is used to represent the abnormal condition of hydrological drought. Thus, the monthly runoff (> 30 years) records matrix is arranged and combined by months (i.e., time series of the monthly average corresponding to year), to obtain the monthly average of runoff. The anomaly series of the monthly runoff in the corresponding month are then calculated (called the anomaly index, *AI*) as follows:

$$AI = \begin{bmatrix} x_{1i} - \bar{x}_1 & x_{2i} - \bar{x}_2 & \cdots & x_{12i} - \bar{x}_{12} \\ x_{1(i+1)} - \bar{x}_1 & x_{2(i+1)} - \bar{x}_2 & \cdots & x_{12(i+1)} - \bar{x}_{12} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1(n)} - \bar{x}_1 & x_{2(n)} - \bar{x}_2 & \cdots & x_{12(n)} - \bar{x}_{12} \end{bmatrix} \quad (1)$$

where 'i' and 'n' represent the first and the last years of the time series; and '1i', '2i', ..., '12i' are 'January', 'February', ..., 'December' in the first year. The rest may be deduced by analogy, '1n', '2n', ..., '12n' represents January, February, ..., December in the last year, and $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_{12}$ represents the monthly average of runoff in a long-term time series (> 30 years).

Because hydrological drought is a cumulative phenomenon resulting from prolonged water shortage (Mckee et al., 1993; Shukla & Wood, 2008), *HAI* and *MAI* use a moving window method for different timescales (e.g., 1, 3, 12, and 24-month timescales). For example, the 1-month moving cumulative window of *AI* for runoff is called *HAI*₁ (the original *AI* sequence in Eq. (1)), the 3-month moving cumulative window of *AI* for runoff is called *HAI*₃, and the successive n-month moving window of *AI* for runoff is called *HAI*_n in this paper. Similarly, the steps for constructing the *MAI* with multiple timescales is the same as that of *HAI*; however, the difference is that the 'x' variable in the Eq. (1) is changed into the precipitation variable, and the [$\bar{x}_1, \bar{x}_2, \dots, \bar{x}_{12}$] is also changed into the long-term average of the precipitation per month.

2.2.2. Determination of thresholds for *HAI* and *MAI* (Step 2 and 3 in Part 1)

Each drought threshold of *HAI* and *MAI* was obtained using the method of equal probability transformation (Lloyd-Hughes & Saunders, 2002). The cumulative distribution frequencies (CDF) of each drought level were obtained from the *SRI* and *SPI* time series of a certain timescale. The thresholds of *HAI* and *MAI* drought levels corresponding to those of *SRI* and *SPI* were then calculated via the equal probability transformation (see Fig. S1). For example, assuming that the threshold of *SRI* (or *SPI*) drought level in a certain timescale is T_1 , the corresponding CDF₁ of hydrological (or meteorological) drought is transformed into the corresponding timescale of CDF₂ for *HAI* (or *MAI*) in an equivalent manner. Thus, the T_2 value of *HAI* (or *MAI*) corresponding to the CDF₂ is a drought threshold of *HAI* (or *MAI*). The original drought levels of *SRI* and *SPI* are 0 according to previous studies (Mckee et al., 1993; Shukla & Wood, 2008) (Step 2 in Part 1). To verify the rationality of *HAI* and *MAI*, the different timescales of *HAI* and *MAI* were compared with the corresponding timescales of the widely used multi-scale standardized runoff and precipitation indices (*SRI*, *SPI*) (Step 3 in Part 1) (Mckee et al., 1993; Shukla & Wood, 2008). We built the *HAI* and *MAI* because they have dimensional features, which help us obtain the amount of water shortage directly during a drought.

2.3. Amount of water required for hydrological drought recovery (Part 2)

The left of Fig. 2 shows a simple example of the watershed yield. According to the water balance of the hydrological cycle for a certain catchment, the flow of a river channel is derived from precipitation, snowmelt, soil moisture, groundwater, or other sources (i.e., import of water into the watershed). In contrast, a decrease in the flow is closely related to evaporation, transpiration, and diversions out of the watershed (e.g., water for human use) (not shown in Fig. 2). There are only two aspects of flow change in the river channel: rising or falling. In other words, when the flow in the river is effectively replenished, the flow will rise, and vice versa. The right part of Fig. 2 (flow process) shows that the river flow rises or falls when the discharges are replenished or weakened. Namely, the flow in the current month minus that in the preceding month is larger than or less than 0. For example, the recharge of flow in a river channel is dependent on precipitation in a rainfall-driven basin. Thus, the amount of runoff required and the corresponding precipitation for the hydrological drought recovery should be obtained. Based on these obtained data, we designed a method to determine the amount of water required (i.e., the amount of runoff to recover from the hydrological drought can be directly obtained from the precipitation) for making typical hydrological droughts (e.g. 2-year, 5-year, and 10-year return periods; referred to herein as 2a, 5a, 10a, respectively) to recover to normal (i.e., the drought index should be larger than the drought threshold), which mainly includes the following two steps.

2.3.1. Describing hydrological drought (Step 1 in Part 2)

- (1) The hydrological drought characteristics are extracted, including drought duration (D), severity (S), and intensity (I) using run theory (the right part of Fig. 2) (Yevjevich, 1967). The drought threshold of *HAI* (or *MAI*) was used to identify the hydrological (or meteorological) drought characteristics, where 'D' is defined as the period from drought start to drought end during which the drought index remains below the truncation level of run theory, 'S' is the cumulative drought index during the D, and 'I' is the average of 'S'.
- (2) The joint return period between D and S is constructed, for example, using bivariate fitting functions (e.g., Gaussian, Clayton, Frank, and Gumbel) in the two-dimensional Archimedean copula family to examine the return period of hydrological drought characteristics (Sklar, 1959). The bivariate fitting functions are used here because drought has multiple features (e.g., D, S, and I).

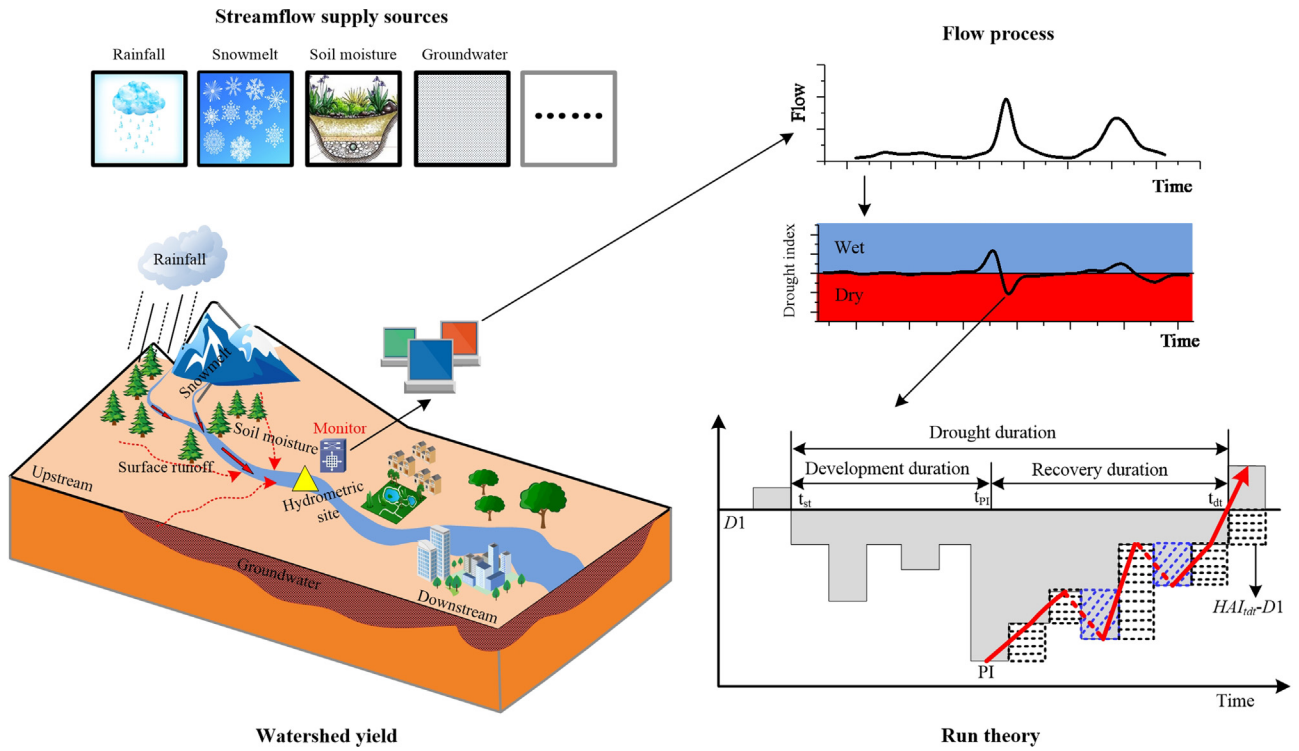


Fig. 2. Dynamic changes in flow at hydrometric stations and possible supply sources in a basin (left); extracting hydrological drought propagation characteristics using the run theory for a certain threshold level (e.g., $D1$) (right); PI is the maximum absolute value of a drought index during the drought duration; the red line (solid and dashed) is the recovery path of hydrological drought in general; the dotted black area is the amount of runoff supplied in the current month relative to the preceding month; the dotted blue area is the amount of runoff shortage for the current month relative to the preceding month. (For interpretation of the references to colour in this figure, the reader is referred to the web version of this article.)

Although D , S , and I are three important properties of droughts, any two of them are sufficient to characterize a single drought (Tu et al., 2018). The parameters for the marginal distribution function and the joint probability distribution function are estimated using the maximum likelihood method, the optimal marginal distribution is selected using the Kolmogorov–Smirnov (K–S) test (Chakravarti et al., 1968), and the optimal joint probability distribution is selected through the root mean square error (RMSE), Nash–Sutcliffe coefficient (N–Sc), and Akaike information criterion (AIC). If the values of RMSE and AIC are small and the value of N–Sc is large, the distance will be minimized between the theoretical and the empirical joint probability distributions of the copula (Borgomeo et al., 2015). Thereafter, the return period of hydrological drought characteristics is determined based on the optimal Archimedean copula function, and the typical droughts in the region are identified. To help the readers better understand the copula function steps, we have described the process in the Supporting Information.

2.3.2. Estimate the amount of water required for recovery (Step 2 in Part 2)

(1) Amount of runoff:

The internal propagation of hydrological drought (Wu et al., 2018a) refers to the process of hydrological drought onset and termination. Onset is defined as the point when the drought index value first decreases such that it is lower (higher) than the drought threshold (e.g., in the run theory of Fig. 2, the $D1$ value is obtained based on the probability of the SPI drought threshold). The hydrological drought development and recovery periods are defined as the periods from onset (the time of drought start, t_{st}) to maximum intensity (PI ; the maximum absolute value of a drought index in the drought), and from PI (the time of PI , t_{PI}) to termination (the time of drought termination, t_{dt}), respectively.

When the difference between the drought index of the current month and that of the preceding months is greater than $D1$, the drought is in the normal recovery phase (denoted by the solid red line in the run theory of Fig. 2). However, the amount of runoff in the river must be replenished during the current month to maintain recovery; otherwise, it is a ‘false recovery’ phenomenon as mentioned above (Wu et al., 2018a). In such scenarios, the monthly runoff would continue to fall, and the drought severity would worsen relative to the preceding month (denoted by the dashed red line in the run theory of Fig. 2). Thus, the amount of runoff required for hydrological drought recovery is defined as the amount of runoff supply (ARS) required during the recovery period. The water supply for the final month of drought recovery (the month before the termination of hydrological drought) is the difference between the final drought index value and $D1$ ($HAI_{dt} - D1$). Therefore, the ARS for recovering hydrological drought is calculated as follows:

$$ARS = (HAI_{dt} - D1) + \sum_{t_{PI}}^{t_{dt}} (HAI_i - HAI_{i-1}) \quad (HAI_i - HAI_{i-1} > 0) \quad (2)$$

where ‘ t_{PI} ’ and ‘ t_{dt} ’ are the times of PI and drought termination, and ‘ i ’ and ‘ $i - 1$ ’ represent the current month and the preceding month during the hydrological drought recovery period, respectively.

(2) Amount of precipitation:

The HAI and MAI values of the current month are affected by the preceding month, we account for this by calculating the variation in the current month’s (i) MAI or HAI value relative to the preceding month’s ($i - 1$) value (i.e. $MAI_i - MAI_{i-1}$, $HAI_i - HAI_{i-1}$) for each month within a given year. Thus, the variable of monthly runoff supply required during the hydrological drought recovery period is input into the relationship of the corresponding month, and the corresponding amount of the monthly precipitation required is obtained. The AP corresponding to the water runoff during the

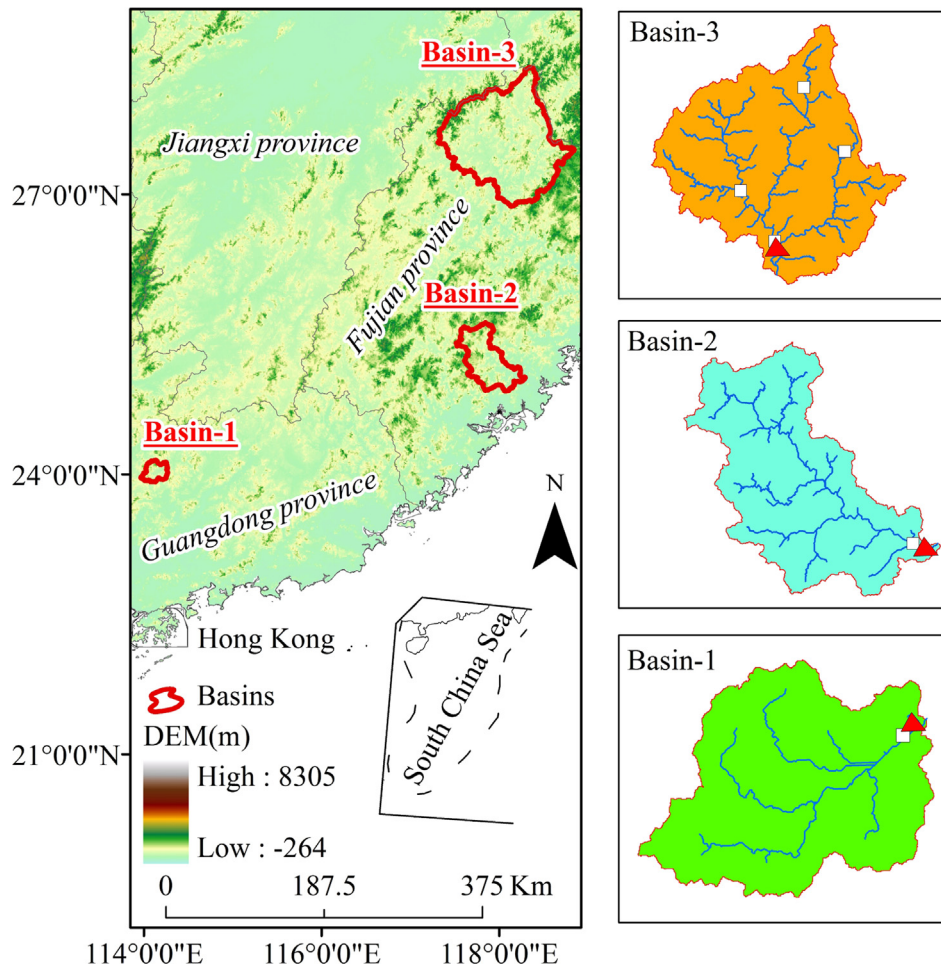


Fig. 3. Map indicating the location of the three river basins. The red triangles indicate the locations of hydrometric stations, while the white squares mark the locations of meteorological stations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

recovery period is regarded as the possible precipitation required for hydrological drought recovery. This is because although the runoff supply for a certain river may be affected by various factors (Van Loon et al., 2016), it is undeniable that precipitation is a key factor driving hydrological drought recovery, particularly in rain-fall-driven basins (Tallaksen and Van Lanen, 2004).

3. Study case and data set

We focused on three rainfall-driven basins in southern China to demonstrate our methods: the Yuecheng (531 km²), the Xixi (2466 km²), and the Jianxi (16,152 km²) river basins (Fig. 3). Hereafter, the basins will be referred to as Basin-1, Basin-2, and Basin-3, respectively. All three basins have unaltered flow (no human regulation via reservoir regulation) and climatology dominated by humid South Asian subtropical monsoons with precipitation and runoff mainly occurring during the warm season (Fig. S2). We used long-term (> 40 years) monthly runoff and precipitation records for each region in the analysis. The monthly runoff and precipitation records from January 1960 to December 2006 for Basin-1 were provided by the Water Conservancy and Electric Power Bureau in Guangdong Province in China. The monthly runoff and precipitation data spanning January 1960 to December 2010 for Basin-2 and from January 1960 to December 2006 for Basin-3 were collected by the Meteorology Agency and Water Conservation Agency of Fujian Province (Table 1). The data underwent strict quality control and have been used for previous hydrological and meteorological drought identification research (Wang

Table 1

The data for this study.

Type	Hydrologic station		Meteorological station	
	Name	Runoff records	Name	Precipitation records
Basin-1	Yuecheng	1960–2006	Yuecheng	1960–2006
Basin-2	Anxi	1960–2010	Anxi	1960–2010
Basin-3	Qilijie	1960–2006	Nan'ping	1960–2006
			Jian'ou	
			Jian'yang	
			Song'xi	
			Pu'cheng	

et al., 2017; Wu et al., 2017, 2018a). The control hydrologic stations for Basin-1, Basin-2, and Basin-3 are Yuecheng, Anxi, and Qilijie, respectively. We used the simple arithmetic mean method to obtain the basin-average monthly precipitation for basins with multiple stations.

4. Results

4.1. Validation of HAI and MAI

The HAI and MAI values for the varying timescales agree with the corresponding SRI and SPI results, particularly for the 12 and 24-month timescales (Figs. S3–S5). As the values of SRI and SPI drop, the values of HAI and MAI are also reduced, and vice versa. In addition, the correlation between HAI (MAI) and SRI (SPI) in the same timescale periods is

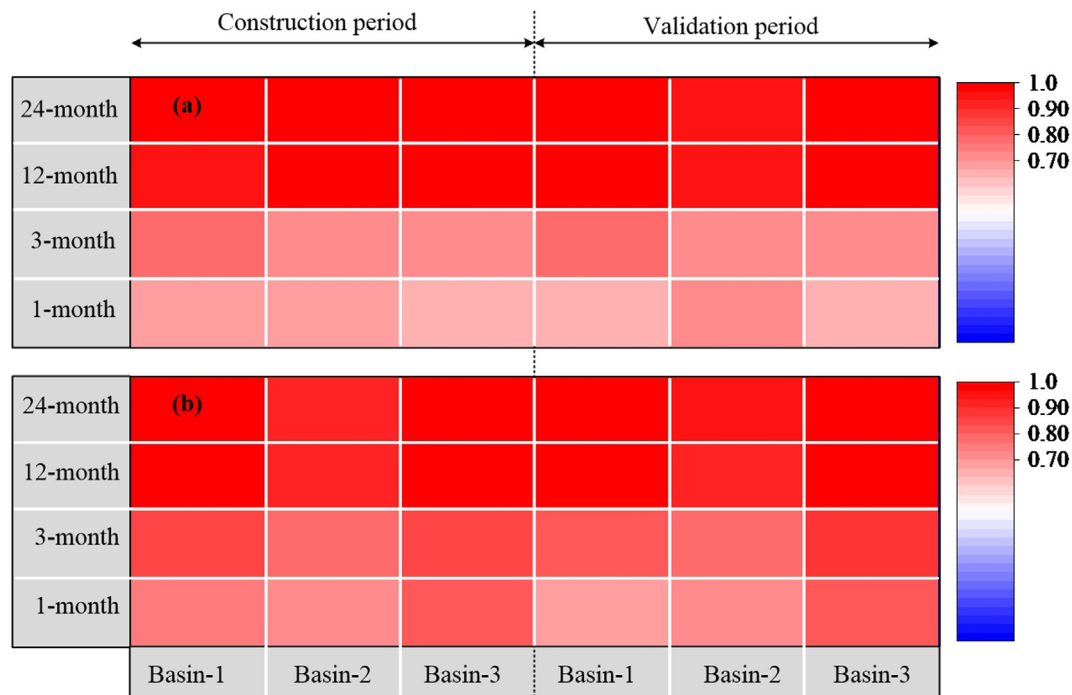


Fig. 4. Correlation among *HAI* and *SRI*, *MAI*, and *SPI* during the construction and validation periods; (a) between *HAI* and *SRI*, (b) between *MAI* and *SPI*, respectively.

relatively high (> 0.70), both in the model development and the validation periods (Fig. 4). This is because the *HAI* and *MAI* are built consistent with *SRI* and *SPI*. Previous studies have also confirmed that *SRI* and *SPI* perform well for identifying hydrological and meteorological drought across the three basins we focus on here (Wang et al., 2017; Wu et al., 2017, 2018a). We further used the 3-month *HAI* and *MAI* (HAI_3 and MAI_3) for drought analysis of our selected study areas due to the timescale of the multi-seasonal droughts in the region (Lu & Wang, 2012). We have included additional details within the Supporting Information that demonstrate the case for obtaining the drought threshold of HAI_3 using the equal frequency transform method (Fig. S1) and a summary of the drought levels of HAI_3 and MAI_3 (Table S1).

4.2. Amount of water required for hydrological drought recovery

4.2.1. Selection of the appropriate marginal distribution and copula function

Following the methods described in Section 2.3.1, we determined that the log-normal distribution function performs better for our study regions relative to the exponential and gamma distributions based on the K-S statistics (Table S2). The $D_{\max}$ for the log-normal distribution is smaller than that of the exponential and log-normal distributions in general. To determine the combined return period of *D* and *S*, four bivariate fitting functions (Gaussian, Clayton, Frank, and Gumbel) were evaluated using goodness-of-fit criteria (Table S3). The minimum values of RMSE and AIC, and the maximum value of $N-Sc$ are Gaussian bivariate distribution functions. Thus, the Gaussian function was selected to determine the joint return period of *D* and *S*, as shown in Fig. 5.

4.2.2. Understanding the typical droughts

Fig. 5 illustrates that there is a representative linear relationship between *D* and *S* within Basin-1 ($R^2 = 0.79$), Basin-2 ($R^2 = 0.91$), and Basin-3 ($R^2 = 0.82$). The drought characteristics samples are mostly within the 95% of the predictive band of the linear relationship. For the three basins, as *D* and *S* increase, the joint return period also shows an increasing trend, and the hydrological *D* and *S* are different in different

return periods, which indicate that the drought prevention and mitigation should be performed based on the differences in hydrological conditions at different hydrometric stations. Additionally, different measures must be taken in other parts of the world for fighting against droughts and reducing disasters.

The typical hydrological droughts (Fig. 5; No.1, No.2, No.3, No.4, and No.5) were selected based on the 2a, 5a, 10a, 25a, and 50a return periods as the longest runoff records have 51 years of data (Basin-2). The selection criteria are proximity to the line (grey dashed line) of the joint return period and the drought samples are within the 95% prediction band of their linear relationship. For Basin-1, they are 1989.08–1990.01 (*D* and *S* are 6 months and $2.62 \times 10^8 \text{ m}^3$), 1977.02–1978.03 (14 months and $5.10 \times 10^8 \text{ m}^3$), 1998.08–1999.09 (14 months and $8.29 \times 10^8 \text{ m}^3$), 1990.06–1992.01 (20 months and $14.46 \times 10^8 \text{ m}^3$), and 2003.05–2005.05 (25 months and $15.93 \times 10^8 \text{ m}^3$), respectively. Meanwhile, for Basin-2, they are 1986.09–1986.12 (4 months and $10.09 \times 10^8 \text{ m}^3$), 1991.02–1991.10 (9 months and $28.35 \times 10^8 \text{ m}^3$), 1967.06–1968.11 (18 months and $48.73 \times 10^8 \text{ m}^3$), 2003.03–2004.10 (20 months and $60.19 \times 10^8 \text{ m}^3$), and 2007.11–2010.01 (27 months and $62.61 \times 10^8 \text{ m}^3$), respectively. Finally, for Basin-3, they are 1980.06–1980.09 (4 months and $58.74 \times 10^8 \text{ m}^3$), 1986.06–1987.03 (10 months and $133.93 \times 10^8 \text{ m}^3$), 1963.02–1963.12 (11 months and $173.51 \times 10^8 \text{ m}^3$), 1995.11–1997.06 (20 months and $291.25 \times 10^8 \text{ m}^3$), and 1971.02–1972.08 (23 months and $361.61 \times 10^8 \text{ m}^3$), respectively.

4.2.3. Water required for hydrological drought recovery in the study basins

For each basin, the Pearson correlation coefficient between HAI_3 and MAI_3 is relatively high (> 0.60) (Fig. 6 (a)). This is true for the current month and for the preceding month (e.g., the higher correlation tends to tilt to the lower right of the 1:1 line). Since *S* of the current month is a cumulative value based on the preceding month, we excluded the effect of *S* in the preceding months and further analyzed the relationship between HAI_3 and MAI_3 . We found that when the cumulative effects are neglected, HAI_3 and MAI_3 agree well for each individual month's drought severity (Fig. 6(b)). Thus, there are 12 linear relationships between the monthly HAI_3 and MAI_3 within a year after eliminating the effects mentioned above for each basin (the detailed

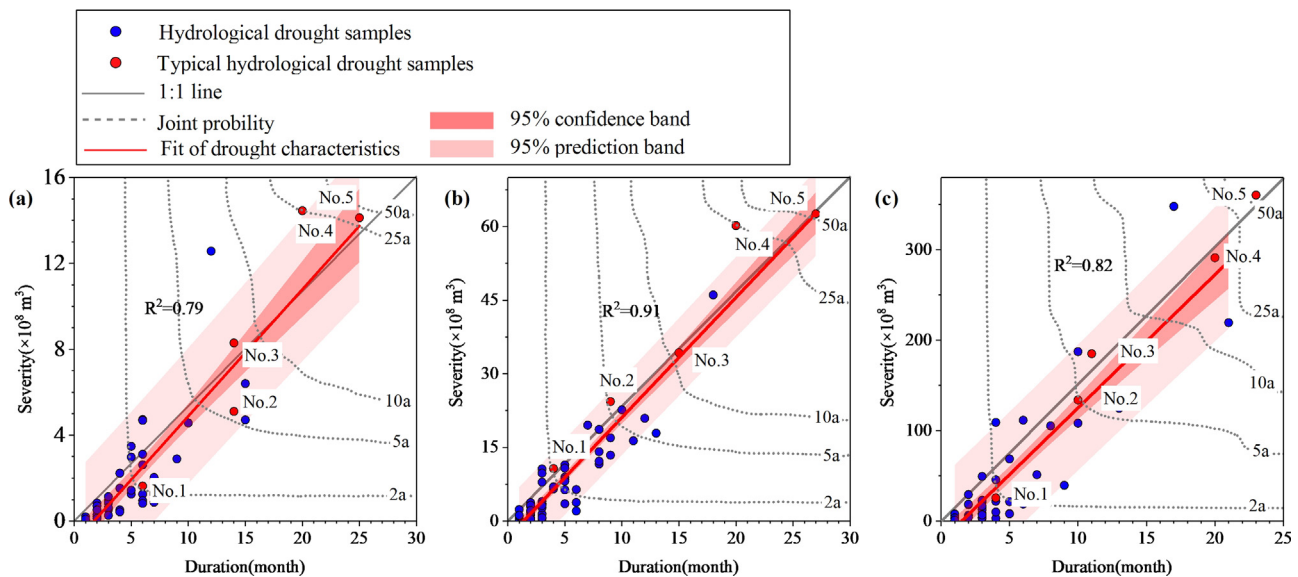


Fig. 5. Linear relationship between drought duration and severity (red line) and their joint return periods (grey dashed line) for Basin-1 (a), Basin-2 (b), and Basin-3 (c). The R^2 is the determination coefficient for the linear relationship. No.1, No.2, No.3, No.4, and No.5 represent the typical hydrological droughts, which are 2a, 5a, 10a, 25a, and 50a, respectively. The red and blue circles are the hydrological drought and typical hydrological drought samples, respectively. (For interpretation of the references to colour in this figure, the reader is referred to the web version of this article.)

linear equation is in Table S4), which are then used to obtain the AP required for recovering from hydrological drought according to the design method in Section 2.3.2 Step 2 (part 2).

As the return period increases (from 2a to 50a), the recovery time for hydrological drought also increases. The average recovery time is from approximately 2 months to 13 months (Fig. 7a). Our results indicate that for Basin-1, the amounts of runoff required for recovery of five typical hydrological droughts (2a, 5a, 10a, 25a, and 50a) are $0.70 \times 10^8 \text{ m}^3$, $1.02 \times 10^8 \text{ m}^3$, $2.31 \times 10^8 \text{ m}^3$, $3.26 \times 10^8 \text{ m}^3$, and $4.25 \times 10^8 \text{ m}^3$, and the corresponding precipitation providing these water quantities are 196.56 mm, 250.69 mm, 544.23 mm, 805.55 mm, and 989.65 mm, respectively. For Basin-2, the amounts of runoff required for five typical hydrological droughts are $4.79 \times 10^8 \text{ m}^3$, $5.79 \times 10^8 \text{ m}^3$, $7.36 \times 10^8 \text{ m}^3$, $10.97 \times 10^8 \text{ m}^3$, and $13.17 \times 10^8 \text{ m}^3$, and their corresponding precipitation amounts are 260.93 mm, 287.32 mm, 387.93 mm, 650.88 mm, and 816.53 mm, respectively. For Basin-3, they are $20.32 \times 10^8 \text{ m}^3$, $26.90 \times 10^8 \text{ m}^3$, $38.71 \times 10^8 \text{ m}^3$, $48.86 \times 10^8 \text{ m}^3$, and $66.25 \times 10^8 \text{ m}^3$, and corresponding precipitation amounts are 116.59 mm, 178.98 mm, 265.95 mm, 351.66 mm, and 499.12 mm, respectively. As detailed in Fig. 7b, there is a strong linear relationship ($R^2 > 0.99$) between the ARS for hydrological drought recovery and the corresponding AP for each of the three basins. These linear equations can then be used to obtain the AP required for a certain hydrological drought in each study area.

5. Discussion

The hydrological and meteorological anomaly indices (HAI and MAI) developed in this study are non-standardized, contain multiple timescale properties, and are better at identifying the propagation process of regional meteorological and hydrological droughts. Because they are calculated based on the back-stepping method of SPI, their fluctuations are consistent with SRI and SPI for the same timescales (Figs. S3–S5), and the correlation between HAI (MAI) and SRI (SPI) is significant (Fig. 4). Hence, they can serve to identify and monitor meteorological and hydrological droughts. Alternatively, the amount of water shortage during hydrological drought propagation can be obtained directly via HAI, whereas SRI and SPI cannot because they provide standardized values. Previous studies (Pan et al., 2013; Antofie et al., 2015) have provided several useful methods for obtaining the AP

required for meteorological drought recovery. However, previous studies cannot estimate the possible precipitation required for hydrological drought recovery. Estimating the required amount of precipitation for drought recovery, however, is highly uncertain. The studies conducted by Pan et al. (2013) and Antofie et al. (2015) were very informative, in which they employed the standardized drought index for analyzing the AP required for meteorological drought recovery. These studies emphasized the importance of considering the drought propagation process (Parry et al., 2016a; Wu et al., 2018a), and not just one specific drought level for a particular month. Moreover, although several recent studies have considered other measures (e.g., water quality and vegetation productivity) for investigating drought recovery (Ahmadi et al., 2019; Campana et al., 2018; Schwalm et al., 2017; Yu et al., 2017), the water required for a certain hydrological drought recovering was not considered. Given the importance of water required for drought recovery for water resource management, we have focused on estimating the amount of precipitation for hydrological drought recovery (Van Loon, 2015).

The results also indicate that the three case studies (Yuecheng, Xixi, and Jianxi river basins) differ in the amounts of runoff and precipitation required for recovering from hydrological drought (Fig. 7), which is related to differences in watershed characteristics (Fig. S1 and Table S1). On the surface, the drought severity in Basin-3 appears greater than in Basin-2 and Basin-1 (Fig. 5); however, the corresponding precipitation required for hydrological drought recovery is the least for the same drought return period. To explain this observation, we further calculate the runoff depth (the ratio between drought severity and catchment area) for the three basins, and we find that the runoff depth of Basin-3 is significantly lower than that of Basin-1 and Basin-2 for the same drought return period, especially the return periods from 5a to 50a (Fig. 8(c)). This shows that different catchment properties can lead to different water requirements for hydrological drought recovery, that is, hydrological drought propagation is not only related to the climate features but is also affected by the catchment characteristics (AghaKouchak et al., 2015b; Booth et al., 2006; Diffenbaugh et al., 2015; Kuriqi et al., 2019; Van Loon et al., 2014, 2016; Wanders et al., 2015).

Although our proposed method allows us to obtain the amount of water required for hydrological drought recovery from a propagation perspective, it still has certain limitations. For example, it is difficult to

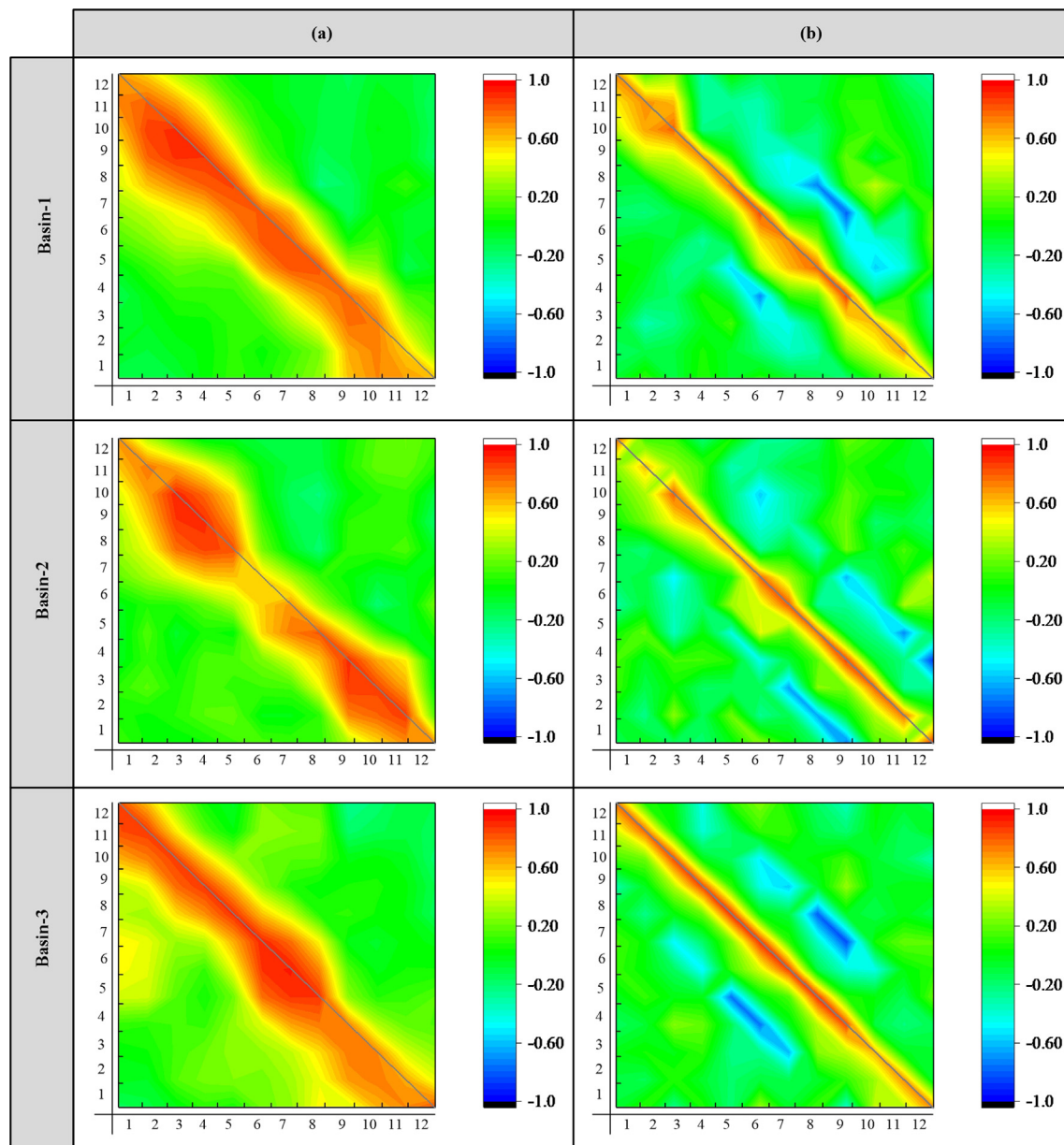


Fig. 6. Correlation between HAI_3 and MAI_3 (a), and the relationship between HAI_3 and MAI_3 (b) when the effect of hydrological (or meteorological) drought severity in the previous months on the drought severity of the current month was excluded in the Basin-1, Basin-2, and Basin-3, respectively.

establish a universal drought index and reasonable drought threshold to monitor and identify all kinds of hydrological drought (Mishra & Singh, 2010). Even though the HAI and MAI with non-standardized features have obvious advantages in obtaining the amount of water required for drought recovery compared to standardized drought indices, it makes it difficult to compare the severity of drought in different regions because they do not provide standardized information. Furthermore, we adopt the 0 threshold of SSI as a case to obtain the HAI drought threshold (D1). Although this threshold is conservative and commonly used, we cannot claim that this threshold works for most applications. Further, we highlight that using different timescales of drought index may lead to different results (Mishra & Singh, 2010). Moreover, hydrological drought recovery is certainly related to rainfall intensity too. How to quantify the impact of rainfall intensity on hydrological drought recovery requires further research. Furthermore, global and regional climate forecasts are increasingly being incorporated for hydrologic forecasting (Zhao et al., 2019, 2020). We plan to integrate a distributed hydrological model with various climate forcings for early warning of

hydrological drought and recovery analysis in our future work.

6. Conclusions

We proposed an approach to determine the amount of runoff required and its corresponding precipitation for a hydrological drought recovery based on drought propagation and non-standardized indices. We demonstrate the application and performance of the proposed method using three rainfall-driven basins with limited human activity located in southern China. The main conclusions can be summarized as follows: Effective management of hydrological drought depends on identification of runoff deficit, reliable estimates of its propagation (development and recovery process), as well as estimates of the water required for its recovery. Additionally, we showed that a linear relationship between the amount of streamflow required and the corresponding precipitation for recovering hydrological drought could be used to obtain the AP required for recovering from a certain hydrological drought. However, different catchment properties cause the

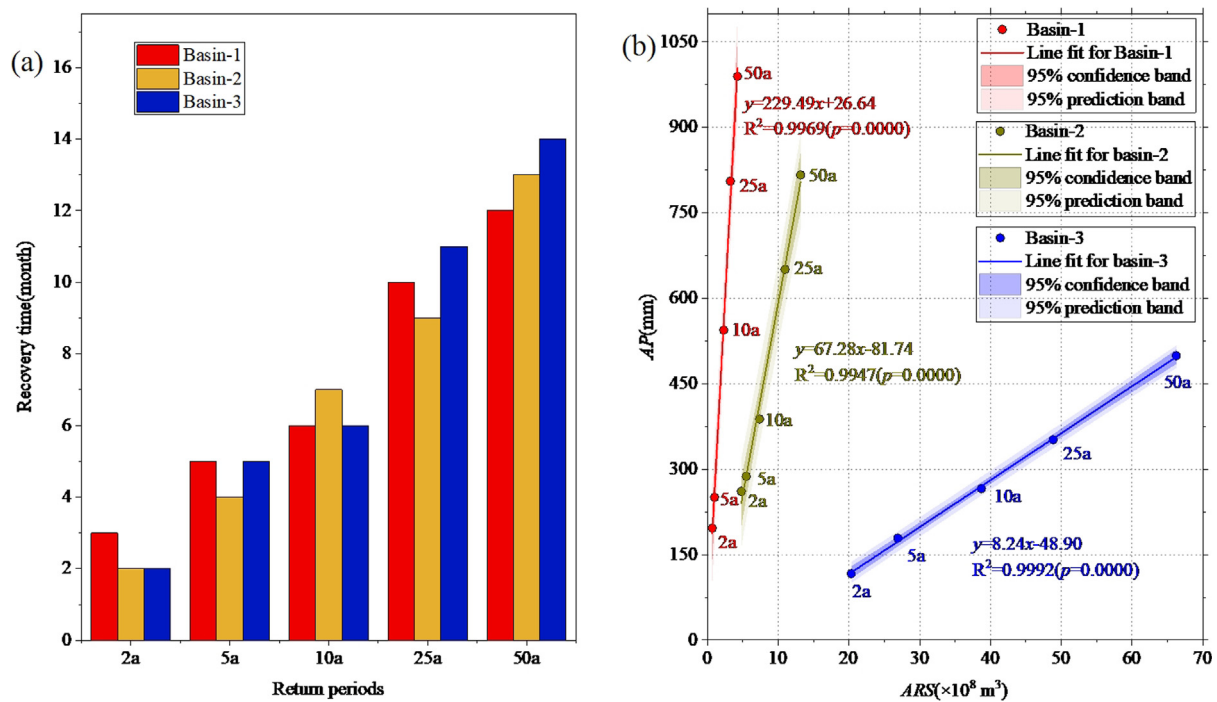


Fig. 7. Hydrological drought recovery time for different return periods (a), and the linear relationship between the amount of runoff required and corresponding precipitation for hydrological drought recovery in a typical hydrological drought (b).

amount of water (runoff and the corresponding precipitation) required for recovering from a hydrological drought to be different, even in basins with similar climatic characteristics. The information about the watershed's natural and man-made characteristics should be considered for hydrological drought early warning, monitoring, and also the water needed for drought recovery.

CRediT authorship contribution statement

Jiefeng Wu: Conceptualization, Methodology, Writing - original draft. Xingwei Chen: Supervision, Project administration, Funding acquisition, Investigation, Data curation. Charlotte Love: Writing - review & editing. Huaxia Yao: . Xiaohong Chen: Data curation. Amir AghaKouchak: .

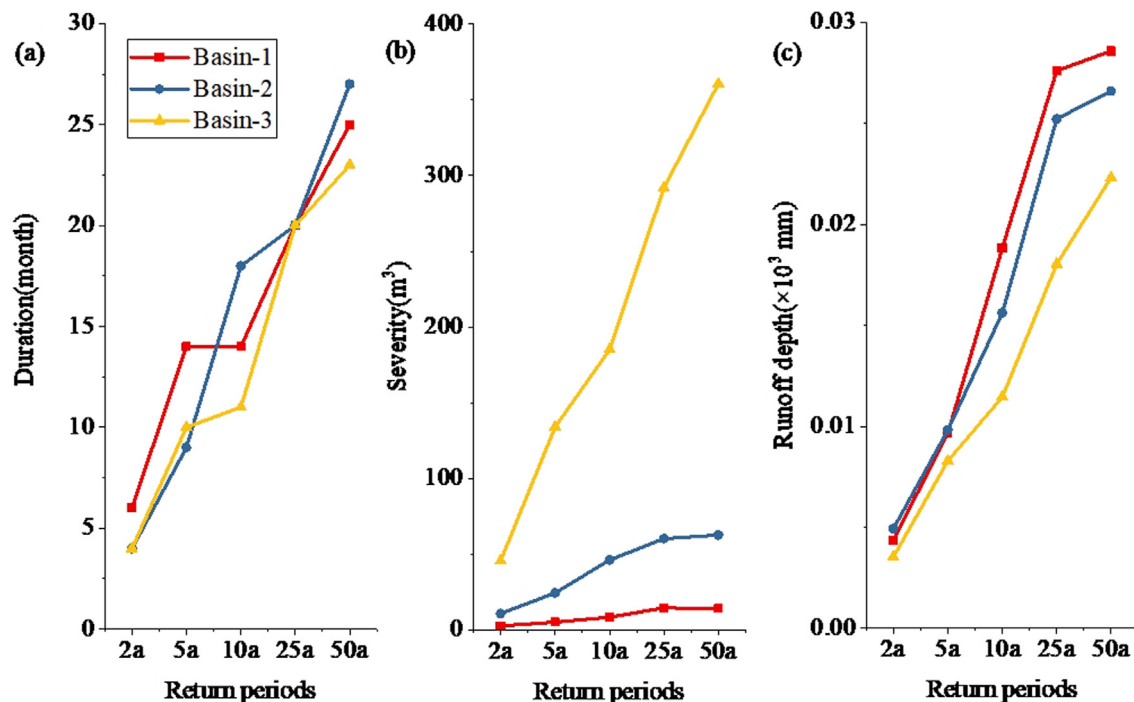


Fig. 8. Comparison of D (a), I (b), and runoff depth during the hydrological drought (c) among three study basins under the same drought return periods. Note that the value of each point in (c) is calculated by the ratio of (b) and (a) points.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2020.125227>.

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